



A satellite image of Jurua river in western Brazil (Courtesy: Planet Labs)

According to Timothy Edgar, visiting fellow, Watson Institute for International Studies, Brown University, USA, "Microsatellites would allow companies, smaller governments or small government entities like state, provincial or local governments, to use overhead imagery with greater resolution and frequency than is typically available today on the commercial satellite imagery market, such as what consumers see with Google Maps. Traditionally, these more powerful images, frequently updated, have been available only to the militaries and intelligence services of major powers. They can be used for military applications and intelligence analysis, and for any number of peaceful purposes such as fighting forest fires, analysing crowds in natural disasters or public emergencies and assisting in law enforcement." He adds, "The number of microsatellites in space could grow quite quickly, as new companies find ways to exploit such imagery. There could be any number of surprising uses, from real estate (commercial and residential) and parking management to corporate security."

*So the new meaning of SaaS could soon be 'satellites as a service!'*

## Smartphones, tablets and, now, satellites

Moore's Law has had its repercussions in more fields than one. As more and more transistors get packed into inte-

grated circuits (ICs), and the chips get increasingly powerful, it is becoming possible to build more powerful yet smaller devices. What computers could do some years ago, smartphones are now able to do with ease.

Satellite technology is one such beneficiary of advanced ICs. Robbie Schingler, co-founder of Planet Labs, a US-based

company, says, "The microsatellite industry benefits greatly from the exponential advances in computing and the increased miniaturisation of consumer electronics. We are piggybacking on decades of research and development in these fields. If you think about what a satellite is, many of the components are that of a supercomputer. The key components of our microsatellite include communication systems (radio, etc), the optics (for imaging), the power system (battery, etc), and the onboard software and computing systems. Our satellites are essentially remote-sensing robots in space, with superfast processors."

"The use of commercial off-the-shelf (COTS) components has been instrumental in the growth of the microsat industry. During the earlier days of satellite building, components were selected, laboriously space-qualified and tested before they were allowed to be used by quality assurance experts, thus pushing the cost high. Digital ICs were ruggedised by shielding them against radiation in space. Today's components are developed to ensure higher reliability. This fact, coupled with the COTS approach, has resulted in the shift towards microsatellites," says John, who was earlier with ISRO. However, he remarks that space-qualified microcontrollers and processors with high processing power are still not readily available, resulting in prohibitively high costs.

Digital IC technology is developing so fast that even nano, pico and femto satellites have become possible. Nanosats weigh between 1 and 10 kg, while pico and femto microsats have wet masses of 0.1 to 1 kg and 10 to 100 gm, respectively.

## All for real, and well within reach

Apart from space research organisations like ISRO, NASA and DARPA, which have successfully launched

## Wanna play space blocks?

CubeSat, developed over a decade ago by professors from Stanford University and California Polytechnic State University to advance satellite design learning, has now become the *de facto* standard for building small satellites—forming the basis of several small satellites being built by NASA, private companies, and start-ups too!

The CubeSat is basically a standardised module, shaped as a cube with 10cm edges and weighing approximately 1.33 kg. It can be fitted with the required instruments and launched into space. A small satellite might consist of one or more CubeSats as building blocks, and it is common practice to describe micro and nano satellites by the number of CubeSat units they contain (e.g., Planet Labs' test satellites sent up in April 2013 were 3U-CubeSats). Generally, satellites may be 1U, 2U, 3U or 6U. CubeSat-based satellites are deployed from a standard poly-picosatellite orbital deployer (P-POD). Endorsing CubeSat's popularity as a building block of research spacecrafts, NASA's CubeSat Launch Initiative (CSLI) provides opportunities for small satellite payloads to fly on rockets planned for upcoming launches. These CubeSats are flown as auxiliary payloads on previously planned missions.

CubeSat.org, an active non-profit organisation, continues to provide the community with not just a standard physical layout, design guidelines and a standard, flight-proven deployment system (P-POD), but also coordination of the required documents and export licences, integration and acceptance testing facilities with formalised schedules, shipment of flight hardware to the launch site and integration to the LV, and confirmation of successful deployment and telemetry information.